

$$1) a - 0,05 + a + 0,2 + 0,15 + 0,3 + 2a = 1$$

$$\Leftrightarrow 0,7 + 3a = 1$$

$$\Leftrightarrow 3a = 0,3$$

$$\Leftrightarrow a = 0,1$$

$$b - i) P(X \geq 3)$$

$$= P(X=3) + P(X=4) + P(X=5)$$

$$= 0,15 + 0,3 + 0,2 = 0,65$$

$$ii) P(X > 3) = P(X=4) + P(X=5)$$

$$= 0,3 + 0,2 = 0,5$$

$$iii) P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= 0,05 + 0,1 + 0,2 + 0,15$$

$$= 0,5$$

c -

$$F_X(c) = \begin{cases} 0 & \text{ne } c < 0 \\ 0,05 & \text{ne } 0 \leq c < 1 \\ 0,15 & \text{ne } 1 \leq c < 2 \\ 0,35 & \text{ne } 2 \leq c < 3 \\ 0,6 & \text{ne } 3 \leq c < 4 \\ 0,8 & \text{ne } 4 \leq c < 5 \\ 1 & \text{ne } c \geq 5 \end{cases}$$

$$d - i) P(X < 2 \mid X \leq 4) = \frac{P(X < 2 \cap X \leq 4)}{P(X \leq 4)}$$

$$= \frac{0,05 + 0,1}{0,05 + 0,1 + 0,2 + 0,15 + 0,3}$$

$$= \frac{0,15}{0,8}$$

$$ii) P(X > 2 \mid X \leq 4) = \frac{P(X=3) + P(X=4)}{0,8}$$

$$= \frac{0,45}{0,8}$$

$$\text{iii) } P(X=4 | X \leq 4) = \frac{0,3}{0,8}$$

$$2) \text{ a - } \int_1^3 kx \, dx = 1$$

$$\Leftrightarrow k \cdot \int_1^3 x \, dx = 1$$

$$\Leftrightarrow k \cdot \left[\frac{x^2}{2} \right]_1^3 = 1$$

$$\Leftrightarrow k \cdot \left(\frac{9}{2} - \frac{1}{2} \right) = 1$$

$$\Leftrightarrow 4k = 1$$

$$\Leftrightarrow k = \frac{1}{4}$$

$$F(x) = \int_1^x \frac{1}{4}t \, dt = \frac{1}{4} \cdot \left[\frac{t^2}{2} \right]_1^x = \frac{1}{4} \cdot \left(\frac{x^2 - 1}{2} \right) = \frac{x^2 - 1}{8}$$

$$F(c) = \begin{cases} 0 & \text{se } c < 1 \\ \frac{x^2 - 1}{8} & \text{se } 1 \leq c \leq 3 \\ 1 & \text{se } c > 3 \end{cases}$$

$$\text{b - i) } P(X > 1,5) = 1 - P(X \leq 1,5) \\ = 1 - \frac{(1,5)^2 - 1}{8} = 1 - \frac{1,25}{8}$$

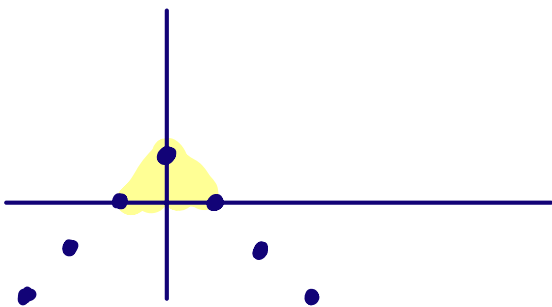
ii) igual

$$\text{iii) } P(1,25 < X < 2) = P(X \leq 2) - P(X \leq 1,25) \\ = \frac{2^2 - 1}{8} - \frac{(1,25)^2 - 1}{8} =$$

$$\begin{aligned}
 &= \frac{3}{8} - \frac{\frac{25}{16} - 1}{8} = \frac{3}{8} - \frac{\frac{9}{16}}{8} \\
 &= \frac{3}{8} - \frac{9}{128} \\
 &= \frac{39}{128}
 \end{aligned}$$

$$\begin{aligned}
 \text{C- } P(x \geq 2 \mid x > 1,5) &= \frac{P(x \geq 2 \cap x > 1,5)}{P(x > 1,5)} \\
 &= \frac{P(x \geq 2)}{P(x > 1,5)} \\
 &= \frac{1 - P(x=2)}{1 - P(x=1,5)} \\
 &= \frac{1 - \frac{3}{8}}{1 - \frac{(\frac{3}{2})^2 - 1}{8}} = \frac{\frac{5}{8}}{1 - \frac{5}{32}} \\
 &= \frac{\frac{5}{8}}{\frac{27}{32}} = \frac{160}{216} = \frac{20}{27}
 \end{aligned}$$

3)



$$a. \quad f(u) = \begin{cases} u+1 & \text{se } -1 < u \leq 0 \\ -u+1 & \text{se } 0 < u \leq 1 \\ 0 & \text{c.c.} \end{cases}$$

$$\text{Para } -1 < u \leq 0$$

$$\int_{-1}^u (t+1) dt = \left[\frac{t^2}{2} + t \right]_{-1}^u = \left(\frac{u^2}{2} + u \right) - \left(\frac{(-1)^2}{2} - 1 \right)$$

$$= \frac{x^2}{2} + x + \frac{1}{2}$$

Para $0 < x \leq 1$

Até $x=0$ acumulamos metade do triângulo Logo $F(0) = 0,5 = \frac{1}{2}$

$$\begin{aligned} F(x) &= \frac{1}{2} + \int_0^x (-t+1)dt = \frac{1}{2} + \left[-\frac{t^2}{2} + t \right]_0^x \\ &= \frac{1}{2} + \left(-\frac{x^2}{2} + x \right) \\ &= -\frac{x^2}{2} + x + \frac{1}{2} \end{aligned}$$

$$F(x) = \begin{cases} 0 & \text{se } x \leq -1 \\ x^2/2 + x + 1/2 & \text{se } -1 < x \leq 0 \\ -x^2/2 + x + 1/2 & \text{se } 0 < x < 1 \\ 1 & \text{se } x \geq 1 \end{cases}$$

b- $P(X=0) = 0$

$$P(X \leq 1/2) = F(1/2)$$

$$= -\frac{1}{8} + 1 = \frac{7}{8}$$

$$P(0 \leq X \leq 1/2) = F(1/2) - F(0)$$

$$= 7/8 - 1/2 = \frac{3}{8}$$

$$P(X \geq 1/2) = 1 - P(X < 1/2)$$

$$= 1 - 7/8 = 1/8$$

$$P(|X| < 1/3) = P(-1/3 < X < 1/3)$$

$$= F(1/3) - F(-1/3)$$

$$= \frac{7}{9} - \frac{2}{9} = \frac{5}{9}$$